Chapter 3 — Bias and fairness

Bias and fairness connect ethical principles to model outcomes once systems affect hiring

Learning objectives

- Define bias in data work
- List five mitigation habits at the ethics-chapter level

What bias means here

- Bias means systematic deviations in collection
- Data are not neutral mirrors
- Example 3.17 states the general risk

Example 3.17 — Bias in AI system

- Example 3.17 — hands-on module
- Example 3.17 highlights domains where errors change life chances, hiring, criminal justice
- Explore the chapter example module
- View files: `modules/chapter3/example17/`

Selection bias

- So error concentrates on underrepresented groups
- Collection design in Chapter 2 is often where skew begins
- Example 3.18 shows a diagnostic model trained mainly on one demographic slice

Example 3.18 — Selection bias in healthcare AI

- Example 3.18 — hands-on module
- Example 3.18 relates to the non-maleficence vignette in Example 3.2
- Explore the chapter example module
- View files: `modules/chapter3/example18/`

Algorithmic bias

- Thresholds can still allocate error unevenly
- Example 3.19 restates hiring rankers that reproduce gender or racial patterns via proxies

Case study pointer — COMPAS

- Example 3.20 points to COMPAS recidivism tools
- The ethics lesson matches Example 3.8, overall accuracy is not a fairness certificate
- Chapter 7 develops detection and impact analysis for this case

Mitigation checklist

- Addressing bias requires better data, explicit fairness criteria
- At this chapter's level

Takeaways

- Fairness work is continuous governance after the principles in Section 3.2
- Selection and algorithmic bias are frequent entry points
- Landmark cases illustrate mechanisms

Next

- Complete the quiz for this part
- Continue to real-world privacy laws, GDPR, HIPAA, CCPA, and selected other regimes